

pH curves for strong acid–base neutralisation reactions

Learning outcomes

By the end of this section you should be able to sketch and interpret the general shape of the pH curve of a strong acid and strong base neutralisation reaction.

Lemon juice, orange juice and grapefruit juice are all acidic. One of the properties of acids is that they have a sour taste. Do all these juices taste as sour as each other? Are they all as acidic as each other? One way to measure the acid content of these juices and other solutions is with titration (**Figure 1**).

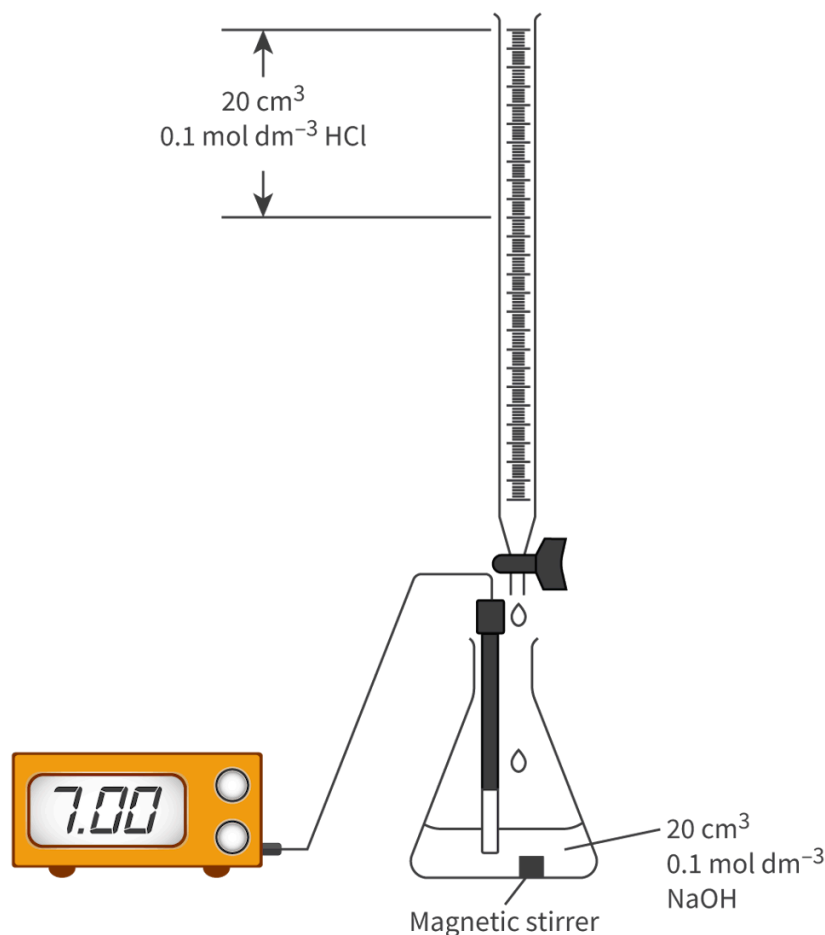


Figure 1. Titration of a strong base (NaOH) with a strong acid (HCl).

 More information for figure 1

The image depicts a titration setup for an experiment involving the titration of a strong base (NaOH) with a strong acid (HCl). The setup includes a tall burette filled with 20 cm^3 of 0.1 mol dm^{-3} hydrochloric acid (HCl), indicated by text near the burette. The acid is gradually being released drop by drop into an Erlenmeyer flask below, which contains 20 cm^3 of 0.1 mol dm^{-3} sodium hydroxide (NaOH). A magnetic stirrer is placed

under the flask to mix the solution for even distribution of reactants. A pH meter is connected to observe the change in pH level, displaying a reading of 7.00, which suggests the solution is neutral at this point of titration. The setup illustrates the controlled environment required for accurate titration experiments.

[Generated by AI]

Titration

Titration is one of the most widely used techniques in quantitative analysis. An acid–base titration is used to determine the unknown concentration of an acidic or basic solution using a solution of known concentration, which is called a standard solution. **Figure 2** shows the apparatus used in an acid–base titration.

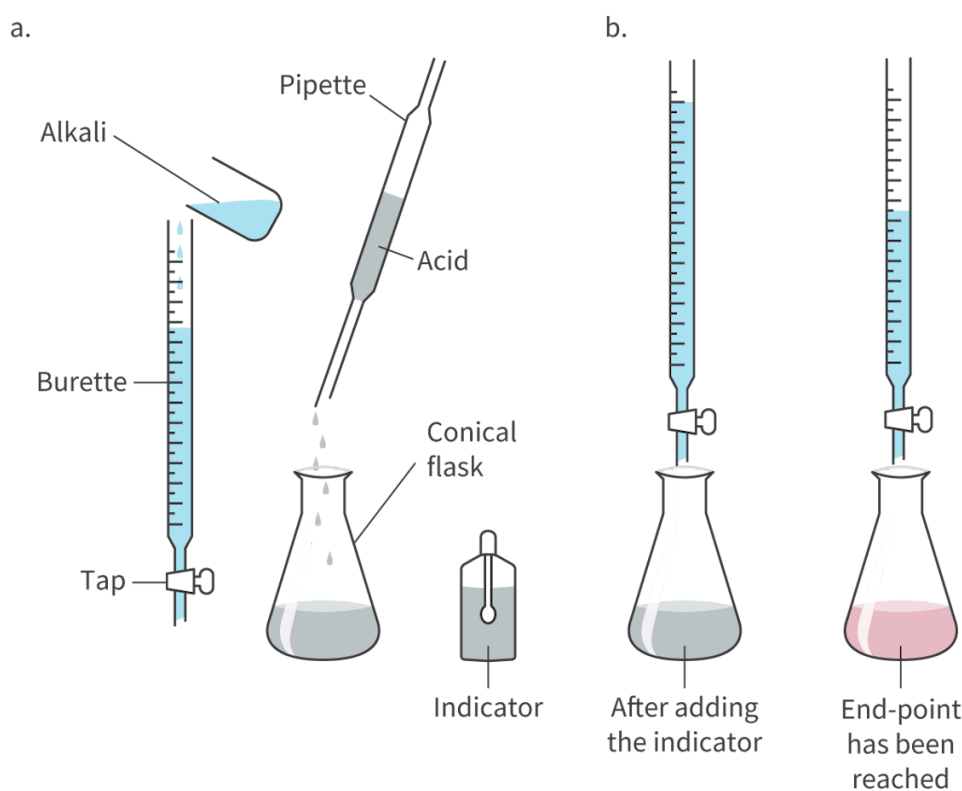


Figure 2. Carrying out a neutralisation by titration.

 More information for figure 2

The image depicts an acid–base titration setup, consisting of two main parts labeled 'a' and 'b.'

In part 'a,' the diagram shows the setup before titration begins: - A burette filled with an alkali solution is positioned vertically on the left side, labeled 'Burette.' The burette has measurement markings, and a tap at the bottom is labeled 'Tap.' - A pipette is held horizontally to transfer an acid solution into a conical flask below. - The conical flask is directly beneath the burette's tip and is labeled. It contains acid and an indicator. - Additional components include a dropper, labeled 'Pipette,' and a diagram of the indicator on the far right, indicating that the color change helps identify the endpoint of the titration.

In part 'b,' the diagram shows two steps of the titration process: 1. On the left, the burette is dispensing the alkali into the conical flask. This step is labeled 'After adding the indicator.' 2. On the right, the titration has reached its endpoint, where the solution in the conical flask has changed color to indicate that the equivalence point has been reached. This is labeled 'End-point has been reached.'

The diagram effectively conveys the procedure and required apparatus for conducting an acid-base titration, highlighting the importance of the visible color change at the endpoint.

[Generated by AI]

The burette is filled with an alkali of known concentration (the titrant). The titrant is added to a measured volume of the acid of unknown concentration (the analyte) until the equivalence point is reached. The equivalence point of the titration is when the acid and base have been reacted in stoichiometrically equivalent amounts. At that point, the solution contains only a salt and water.

The equivalence point is signified when the indicator changes colour. In the example below, the indicator is phenolphthalein. Note that the colour of phenolphthalein in an alkaline solution can be described as pink/purple and can fade to colourless in strong alkali if left to stand. The volume of the standard solution is measured.

Making connections

You studied stoichiometry calculations from subtopic S1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/>)? Sometimes reactants are given in excess, sometimes they react completely in the correct ratio, with none in excess. Can you think why is the equivalence point sometimes referred to as the stoichiometric point?

This concept can be applied when testing food components to see if they have too much, too little or just the correct amount of a component.

For example, the content of ascorbic acid in fruit juices in order to compare and determine if the fruit juice has advertised levels of ascorbic acid can be determined by titration. You can explore this [site](https://www.mt.com/in/en/home/applications/laboratory/food-and-beverages/acidity-measurement.html)  (<https://www.mt.com/in/en/home/applications/laboratory/food-and-beverages/acidity-measurement.html>) for more information.

pH curve

In a neutralisation reaction between an acid and a base, there will be a change in pH. This depends on whether the acid is added to the base or vice versa. When a strong acid is added to a strong base, the pH of the solution (as measured by a pH probe) will decrease from a high pH (approximately 14) to a lower pH as the

concentration of H^+ ions increases. The resulting graph showing the change in pH is known as a pH curve. It shows the change in pH vs. the volume of acid or base being added.

In the titration of HCl with NaOH, a student recorded the pH and volume values in **Table 1**

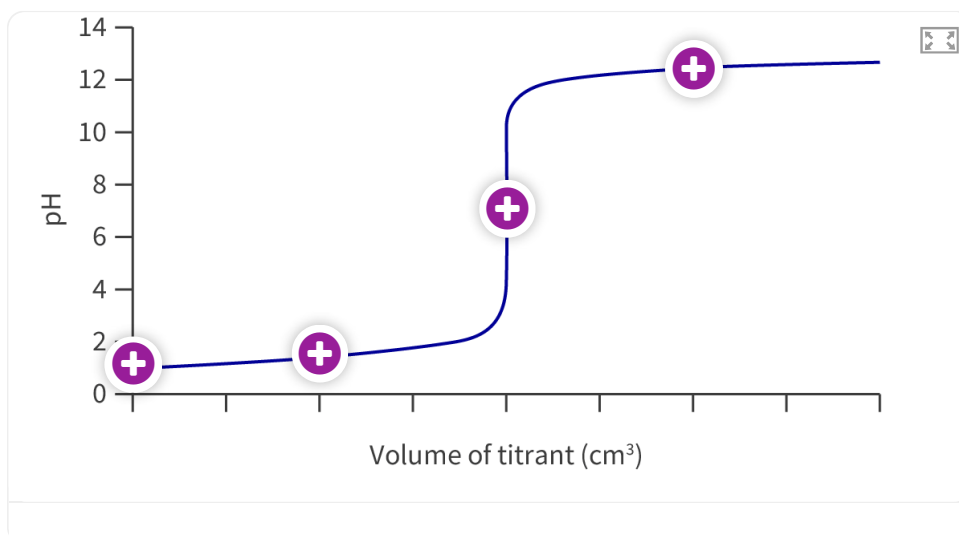
Table 1. Student results for titration of HCl with NaOH

pH	Volume of NaOH added (cm^3)
3.61	0.00
3.92	3.00
4.13	6.00
4.29	9.00
4.43	12.00
4.45	15.00
4.96	18.00
5.00	21.00
5.12	24.00
5.28	27.00
5.46	36.00
5.73	39.00
6.29	42.00
10.84	45.00
11.67	48.00
11.93	51.00

pH	Volume of NaOH added (cm ³)
12.03	54.00
12.18	57.00

Try plotting the data. Can you describe key features of the graph? What can you say about the concentration of acid and base at the equivalence point?

Use **Interactive 1** to investigate the main attributes of a typical pH curve from a titration.



Interactive 1. A Typical pH Curve From a Titration.

 More information for interactive 1

An interactive graph plots pH on the vertical axis, ranging from 0 to 14, and the volume of titrant (cm³) on the horizontal axis. The graph displays an S-shaped curve with four plus icons each representing a hotspot.

The hotspots are named Hotspot 1, Hotspot 2, Hotspot 3, and Hotspot 4 from right to left. The hotspot 1 is located on the curve where the pH is approximately 12. The hotspot 2 is located on the curve where the pH is approximately 7. The hotspot 3 is located on the curve where the pH is approximately 2. The hotspot 4 is located on the curve where the pH is approximately 1.

Clicking on the hotspots reveals the following details:

Hotspot 1:

The curve ends at a high pH (pH of a strong base) since there is now an excess of strong base that ionises completely.

Hotspot 2:

The equivalence point can be determined by finding the point half-way up the steepest part of the curve and drawing a horizontal line across to the y-axis. For the titration of a strong acid and a strong base, the pH at the equivalence point is 7. At the equivalence point, the acid and base have been reacted in stoichiometrically equivalent amounts and the solution contains only salt and water.

Hotspot 3:

With the addition of a strong base to neutralise the acid, the pH gradually increases until it undergoes a sharp increase between pH 3 and pH 11; this is known as the point of inflection.

Hotspot 4:

The pH curve starts at a low pH (pH of a strong acid) since acid ionises completely. This interactive graph helps users understand how the pH changes during a strong acid–strong base titration.

Study skills

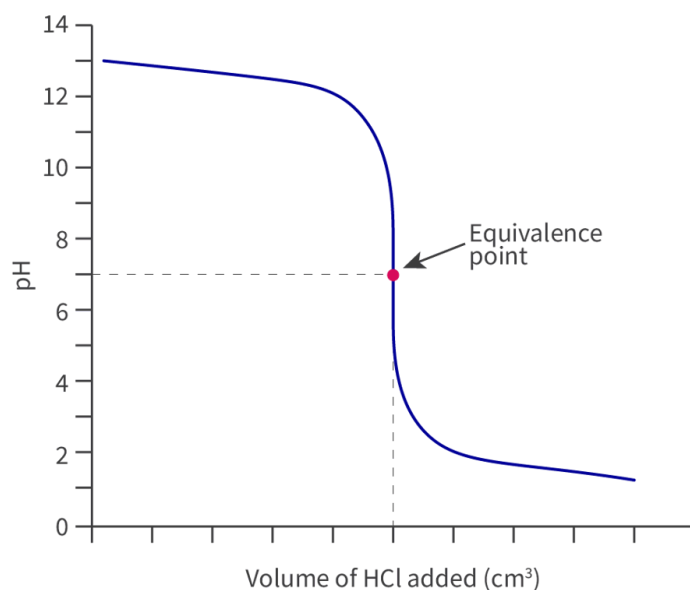
Interpretation of the graph only should include the intercept with the pH axis and equivalence point.

Only monoprotic neutralisation reactions will be assessed.

Worked example 1

Sketch a titration curve for the titration of NaOH with HCl in which the acid is added to the alkali.

Label each section of the graph.



1. The pH curve starts at a high pH (pH of a strong base).
2. With the addition of a strong acid, the pH gradually decreases until it undergoes a sharp decrease between pH 11 and pH 3; this is known as the point of inflection.

3. The equivalence point can be determined by finding the point half-way up the steepest part of the curve and drawing a horizontal line across to the y -axis.
4. For the titration of a strong acid and a strong base, the pH at the equivalence point is 7.
5. At the equivalence point, the acid and base have been reacted in stoichiometrically equivalent amounts and the solution contains only salt and water.
6. The curve ends at a low pH (pH of a strong acid).

Practical skills

Tool 1: Experimental techniques — Applying techniques

Tool 3: Technology — Applying technology to process data

Inquiry 2: Collecting and processing data — Interpreting results, interpret diagrams, graphs and charts

How can titration be used to calculate the concentration of an acid or base in solution?

Think about [subtopic S1.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/) and stoichiometry when you perform the acid–base titration in the lab

Remember that when you graph your pH curve, you will use the graph as described above to determine the equivalence point and you will use the information obtained to perform the required calculations.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills— Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

Instructions

You are given the titration results in **Table 1**.

Table 1. Titration results.

Volume of NaOH (cm ³)	
0.00	
2.00	
4.00	
6.00	
8.00	
10.00	
12.00	
14.00	
16.00	
18.00	
20.00	
22.00	
24.00	
1. Use the data to sketch a titration curve. Make sure to clearly label the axis, showing the data points and line connecting them. 2. Label the various parts of the graph describing the stages of the titration. 3. Determine the pH at the equivalence point.	